# Introduction to Social Data Analytics: Lecture 8 Part 1

# Today's Lecture

- Introduce the empirical application of the week: Greenstone et al. (2022)
- Discuss how to make an assortment of figures in base R
  - histograms: how does pollution vary across cities in China?
  - box plots: what is a boxplot and how can we use it to study changes in pollution over time?

# Application: Air Pollution Monitoring

- Greenstone et al. (2022) examine the introduction of automatic air pollution monitoring in China
  - Local officials were given incentives to meet pollution goals
  - But they also had the potential to manipulation local pollution data
- Greenstone et al. (2022) study how reported levels of PM10 change after automated monitoring of pollution levels
- PM10 is a measure of particulates in the air that are 10 micrometers in diameter or smaller

# Application: Air Pollution Monitoring

- In R, we can load Stata datasets in a couple of ways
- One way is through the haven package
- Load data:

```
library(haven)
library(tidyverse)
```

```
## Warning: package 'tibble' was built under R version 4.1.2
```

```
pm <- read_dta("station_day_1116.dta")
pm <- filter(pm, !is.na(pm10))
```

- The final line filters out observations with missing pollution data

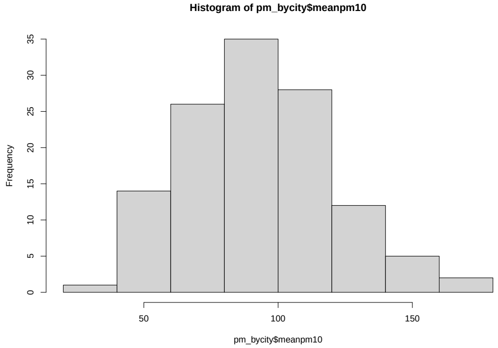
## Group by city:

```r
#Get average pm10 by city
pm_bycity <- pm %>%
  group_by(code_city) %>%
  summarize(meanpm10 = mean(pm10, na.rm=T))
#Look at the data
head(pm_bycity)
```

```
## # A tibble: 6 x 2
##   code_city meanpm10
##       <dbl>    <dbl>
## 1    110100    110.
## 2    120100    122.
## 3    130100    166.
## 4    130200    132.
## 5    130300     92.2
## 6    130400    145.
```

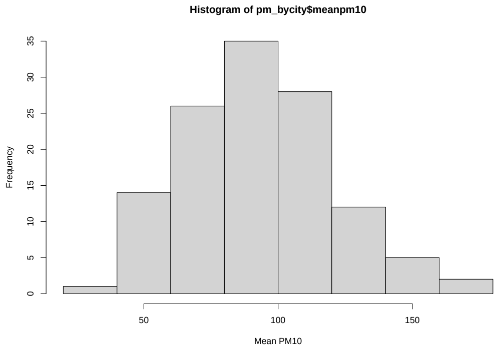
# Create a basic histogram

```
hist(pm_bycity$meanpm10)
```



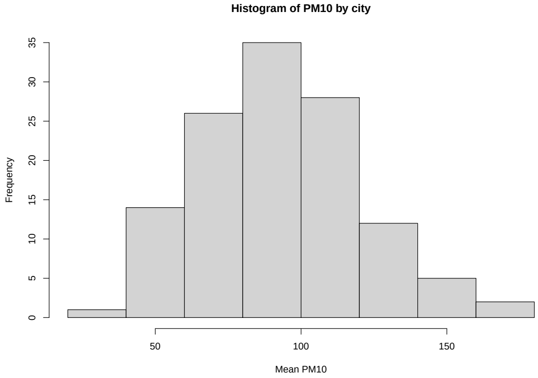
# Add x and y axis labels

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",
     ylab="Frequency")
```



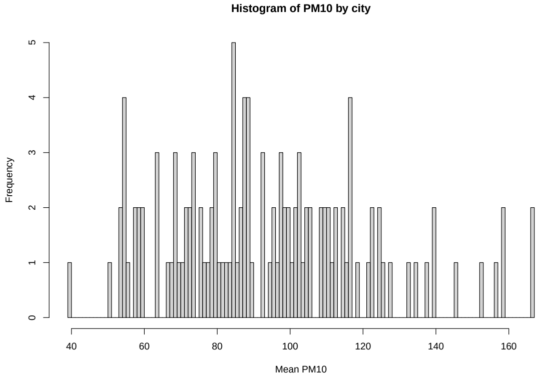
# Add title

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",
  ylab="Frequency",
  main="Histogram of PM10 by city")
```



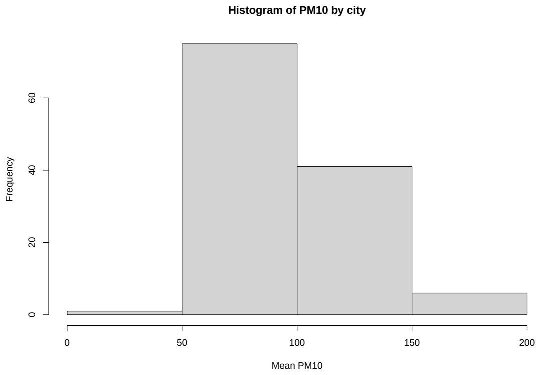
# More breaks

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",
  ylab="Frequency",
  main="Histogram of PM10 by city", breaks=100)
```



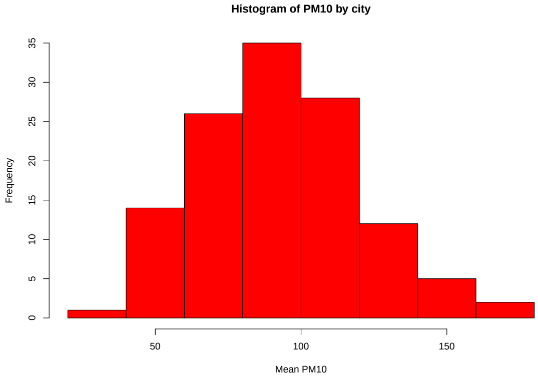
# Fewer breaks

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",
  ylab="Frequency",
  main="Histogram of PM10 by city", breaks=4)
```



# Add color

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",
  ylab="Frequency",
  main="Histogram of PM10 by city", col="red")
```




- For more colors in R: http://www.stat.columbia.edu/~tzheng/files/Rcolor.pdf

# Comparing Histograms

- As part of our analysis, we are going to compare two histograms, one before and one after automation

# Split data into before and after automation

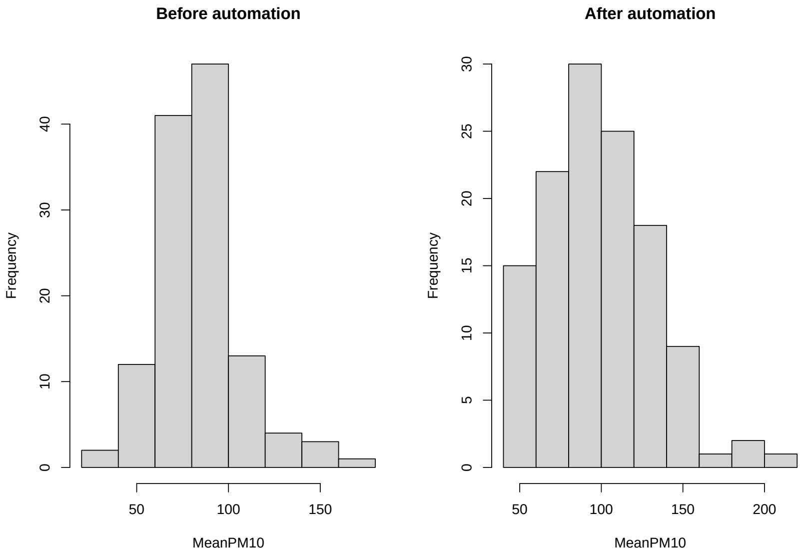
```
#Create data for before automation
pm_bycitybefore <- pm %>%
  mutate(T=date - auto_date) %>%
  filter(T< 0) %>%
  group_by(code_city) %>%
  summarize(meanpm10 = mean(pm10))

#Create data for after automation
pm_bycityafter <- pm %>%
  mutate(T=date-auto_date) %>%
  filter(T > 0) %>%
  group_by(code_city) %>%
  summarize(meanpm10 = mean(pm10))
```

# Create two plots

```r
#Create two panes for plots
par(mfrow=c(1,2))
#Plot histogram for before
hist(pm_bycitybefore$meanpm10, xlab="MeanPM10",
  ylab="Frequency",
  main="Before automation")
#Plot histogram for after
hist(pm_bycityafter$meanpm10, xlab="MeanPM10",
  ylab="Frequency",
  main="After automation")
```

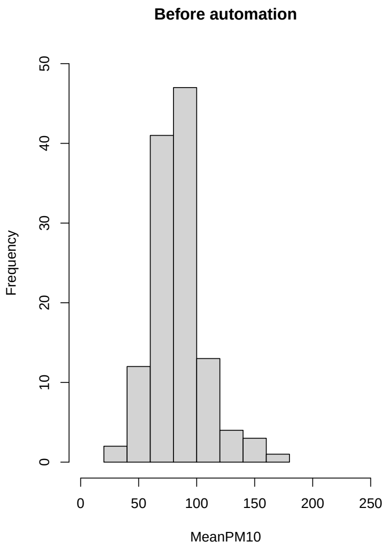
# Create two plots

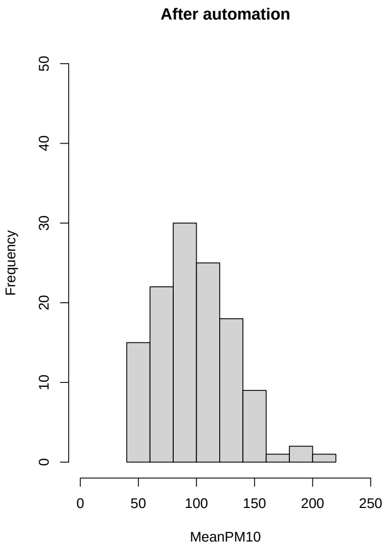


# Change x- and y-axis limit to make them comparable

```r
#Create two panes for plots
par(mfrow=c(1,2))
#Plot histogram for before
hist(pm_bycitybefore$meanpm10, xlab="MeanPM10",
  ylab="Frequency",
  main="Before automation",
  xlim=c(0,250), ylim=c(0,50))
#Plot histogram for after
hist(pm_bycityafter$meanpm10, xlab="MeanPM10",
  ylab="Frequency",
  main="After automation",
  xlim=c(0,250), ylim=c(0,50))
```

# Change x- and y-axis limit to make them comparable



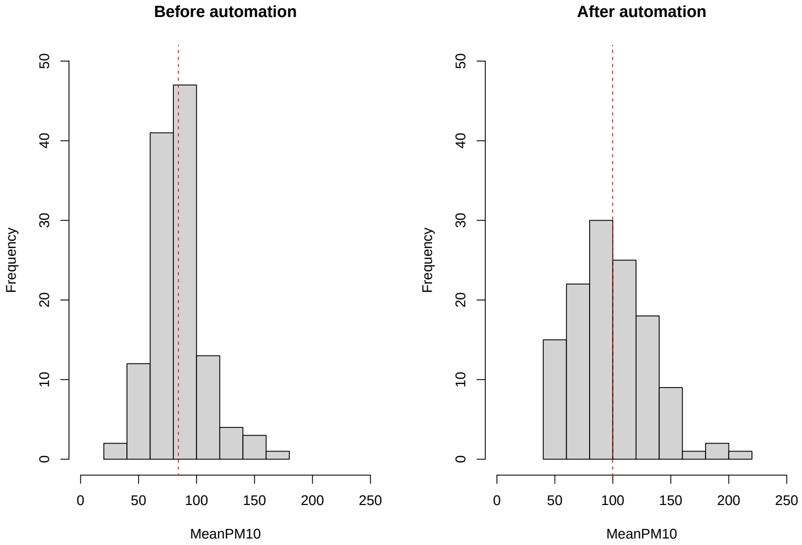
# Adding lines to a plot

- We can use the function `lines()` to add a line to the plot
- To do so we must specify $(x_1, x_2)$ and $(y_1, y_2)$. The line starts at the point $(x_1, y_1)$ and ends at $(x_2, y_2)$.
- In our graph, if we want to place vertical line at the mean we can specify `lines(c(mean,mean),c(-10,100))`, where `mean` is an object that stores the mean value of pollution
  - In this example the vertical axis goes from 0 to 50. By specifying `c(-10,100)` the line will span the entire graph.

# Add lines to the plot

```
meanbefore <- mean(pm_bycitybefore$meanpm10)
meanafter <- mean(pm_bycityafter$meanpm10)
#Create two panes for plots
par(mfrow=c(1,2))
#Plot histogram for before
hist(pm_bycitybefore$meanpm10, xlab="MeanPM10",
  ylab="Frequency",
  main="Before automation",
  xlim=c(0,250), ylim=c(0,50))
lines(c(meanbefore, meanbefore), c(-10, 100),
  lty=2, col="red")
#Plot histogram for after
hist(pm_bycityafter$meanpm10, xlab="MeanPM10",
  ylab="Frequency",
  main="After automation",
  xlim=c(0,250), ylim=c(0,50))
lines(c(meanafter, meanafter), c(-10, 100),
  lty=2, col="red")
```

# Add lines to the plot



# Boxplot

- A box plot is a figure we have not plotted yet
  - It displays a few relevant statistics about a variable of interest, such as the median, 25th percentile, and 75th percentile
  - It is useful when you want to compare these statistics across groups
- In our case, we want to understand the difference in these statistics before and after automation

# Group air pollution by time to automation

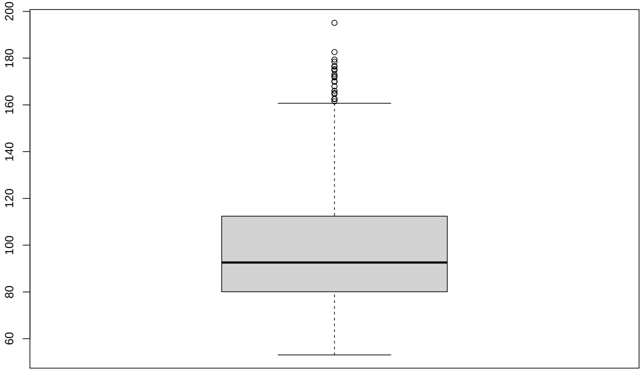
```
#Group by day
pm_byday <- pm %>%
  mutate(
    T=date-auto_date) %>%
  filter(T>-364, T<364) %>%
  group_by(T) %>%
  summarize(meanpm10 = mean(pm10))
```

# Make a boxplot for mean PM10

- Middle line of boxplot is median
- Upper and lower of box are third and first quartile
- Inter-quartile range (IQR): length of box
- Upper whisker: Add $1.5 * IQR$, or maximum
- Lower whisker: Add $1.5 * IQR$, or minimum

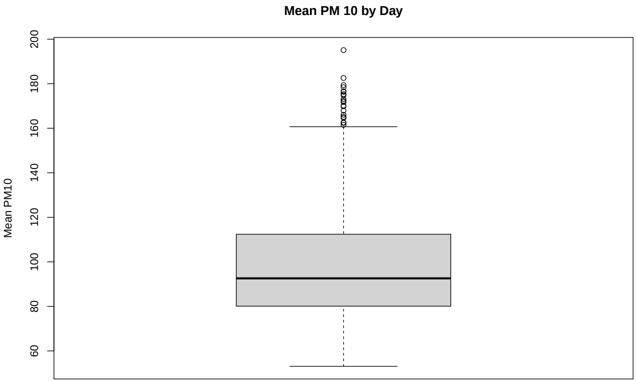
```
#Create Boxplot
boxplot(pm_byday$meanpm10)
```

# Make a boxplot for mean PM10



# Customize the boxplot

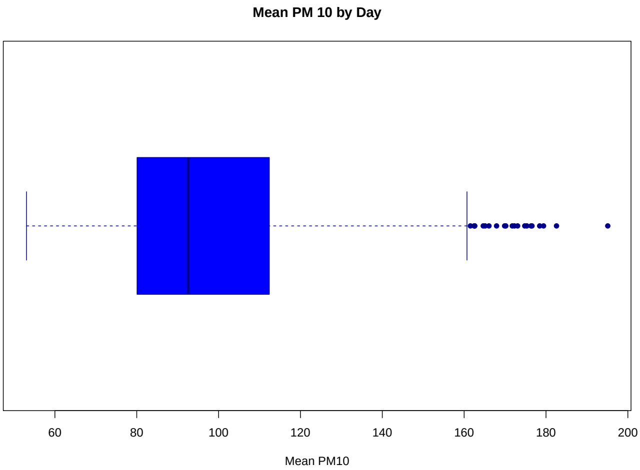
```
#Create Boxplot
boxplot(pm_byday$meanpm10, ylab="Mean PM10",
  main="Mean PM 10 by Day")
```



# Customize the boxplot

```r
#Change color, flip horizontal
boxplot(pm_byday$meanpm10, xlab="Mean PM10",
main="Mean PM 10 by Day",
  col="blue", border="darkblue",
  pch=16, horizontal=T)
```

# Customize the boxplot



# Plot by another variable

```r
#Create after variable
pm_byday$after <- pm_byday$T>0
#Split into two boxplots
boxplot(pm_byday$meanpm10 ~ pm_byday$after, xlab="Mean PM10",
  main="Mean PM 10 by city", col="blue", border="darkblue",
  pch=16, horizontal=T, ylab="After Automation")
```

# Plot by another variable